



Awesense Energy Transition Platform Use Cases

● - demoable implementation

🔓 - demoable & open sourced implementation

Apr 28, 2026



Jupyter



Power BI



Has Sandpit
Dataset to try?

UC01	Coincident Peak and Non-Coincident Peak Analysis	🔓		●
UC02	Power Factor Analysis		●	
UC03	Phase Balance Analysis		●	●
UC04	Transformer Capacity Analysis for EV Charging	🔓		●
UC05	Reversed Power Flow and PV Capacity Analysis	🔓		●
UC06	GIS Grid Connectivity Validation and Correction		●	●
UC07	Asset Management - Cables and Lines			
UC08	Asset Management - Substations			
UC09	EV Charging and Use of Reserved Capacity Analysis	🔓		●
UC10	Connected Loads [kW] vs. Actual Phase Balance Analysis			
UC11	Analysis of Unserved Energy Due to Outage			
UC12	Grid Temperature Monitoring		●	●
UC13	Master Circuit Breaker Value vs. Measured Values Analysis	🔓		●
UC14	Onsite Generation and Ghost Load Analysis			
UC15	Utility Data Challenge Hub		●	●
UC16	Registered PV Generation Analysis	🔓		●
UC17	Actual vs. Contracted Grid Capacity Analysis	🔓		●
UC18	Grid Alerts and Warnings		●	●
UC19	IoT Health Index and Overview		●	
UC20	Unexpected Generation Identification			
UC21	Geospatial Analysis of Grid Alerts and Warnings		●	●



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UC22	Topological Grid Segmentation			●	●
UC23	Analysis of Grid Segments and Elements			●	●
UC24	Analysis of Planned Outage for Assets Upgrades	🔗	●		●
UC25	Analysis for Maximizing Value of IoT and Meter Devices			●	
UC26	Transformer Loading Analysis	●			●
UC27	IoT Sensor Billing	●		●	●
UC28	Energy Balancing for Revenue Protection			●	●
UC29	Non-Technical Loss Investigations Management			●	●
UC30	Suspicious Pattern Identification and Analysis			●	●
UC31	Grid and Time Series Visualization			●	●
UC32	Power Outage Visualization			●	
UC33	MV Grid Optimization Analysis				
UC34	EV and Appliance Disaggregation				
UC35	Standard Consumption Patterns Creation				
UC36	Phase Identification Analysis				
UC37	Detection of Meter Incorrect Wiring Connection				
UC38	Transformer Load Increase Assessment Tool	🔗			●
UC39	IPT and ICT Verification for Proper Billing			●	
UC40	Phase Imbalance Correction Suggester	●			●
UC41	EV Load Growth Forecasting	🔗			●
UC42	Meter to Transformer Association Analysis				



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UC43 Load Forecasting

UC44 Voltage Power Quality Monitoring

UC45 Feeder Voltage Analysis and Visualisation

UC46 Feeder Capacity and Lateral Capacity Analysis

UC47 Dynamic Line Rating Analysis

UC48 AMI-Smart Meter Flag Analysis for Theft Detection

UC49 Time Of Use Rate Analysis

UC50 Work Orders Management

UC51 Wildfire Risk Assessment

UC52 Hosting Capacity Analysis

UC53 Grid Map Publication Tool

UC54 Network Model Diff Tool

UC55 Equipment and Asset Type Catalogue

UC56 Asset Inspection Data Quality and Anomaly Detection

UC57 Reliability Metrics Trending and Benchmarking

UC58 Asset Maintenance History and Inspection Records Analysis

[illegible]

[illegible]

